YEAR 8 · SCIENCE DETECTIVES 科学侦探 · WEEK 2 OF 10 · 40 MINUTES

Fingerprints 指纹档案

Scientists classify things and count how common each group is; how common a feature is decides how strong it is as evidence.

- I can take my own fingerprint and classify it as a loop, whorl or arch.
- I can use class data in a bar chart to explain why a common fingerprint type is weak evidence.

fingerprint 指纹 · **pattern** 纹型 · **loop** 箕形纹 · **whorl** 斗形纹 · **arch** 弓形纹 · **classify** 分类 · **tally** 划记计数 · **bar chart** 条形图 · **common / rare** 常见的／罕见的

THE LESSON

Do Now	4	Four questions from the suspect cards; check roles; log row 1 if not done; Today slide
Hook	3	Fingerprint facts and an ancient Chinese finger-mark contract; fingertips under magnifiers
Learn	7	Loop, whorl, arch with gestures; how common each is; teacher demos the pencil-and-tape method
Do	18	Take prints (5 min), classify (3), tally and bar chart (6), Fingerprint File at minute 14 (4)
Think and Reflect	8	How strong is the fingerprint evidence? Pair talk; evidence log row 2, tracker; exit ticket

Hands-on: Students take own prints with soft pencil and tape onto a record card, classify with a magnifier, add a tally.

Collect or assess: Record card; tally and bar chart; evidence log row 2 and suspect tracker; exit ticket on common vs rare evidence.

SAFETY

- Graphite on fingers: wash hands after taking fingerprints; wipe tables.
- No pencils or tape in mouths.
- Take only your own fingerprints.

WATCH OUT

- A matching type is not proof: the Think answer is 'sometimes'; 3 of 4 suspects have loops.
- Smudges: lots of graphite, gentle press, no sliding. One tally per student, right index finger only.

Language: Frames on tables; Chinese talk and answers welcome. Band 1 beside a Band 3 buddy. Traps: 'loop', 'arch', 'tally'.

Next week: Week 3 (chromatography): keep tape rolls; this week cut filter strips 2 cm by 10 cm, make pen B ink 'N' strips

PREP FOR ONE CLASS

before the lesson about 50 min (35 could be done by someone else); after about 21 min (some times estimated)

2 OR MORE WEEKS BEFORE

- [ ] Order pencils 2B to 4B x24, clear tape x6 rolls, A4 white card x12 sheets 10 min · technician
- [ ] Book printing: worksheets x24 (48 sheets), kit card x1 (A5) 5 min · reprographics

THE WEEK BEFORE

- [ ] Gather prep room kit; check visualiser works and sink has soap 10 min · technician
- [ ] Print record cards on white card (12 sheets, cut) and Fingerprint File x7 10 min · technician
- [ ] Check each case file has its suspect cards and evidence log 5 min · teacher
- [ ] Practise one pencil-and-tape print before the demo (optional) optional 5 min · teacher

THE DAY BEFORE

- [ ] Plan seating: each Band 1 student beside a Band 3 language buddy 5 min · teacher
- [ ] Tape scene print onto a drinking glass for display (optional) optional 2 min · technician

ON THE DAY, BEFORE THE LESSON

- [ ] Draw tally table as Slide 9; set out 6 trays, table kit; visualiser on 5 min · teacher

AFTER THE LESSON

- [ ] Clean-up: cards into case files, kit in trays, wipe tables, wash hands 3 min · students
- [ ] Read exit tickets, check some record cards; files, logbooks to class box 10 min · teacher
- [ ] Between classes: count pencils and tape; refill trays; replace marked Files 5 min · technician
- [ ] After last class: return magnifiers, trays, cloths; keep 6 tape rolls 3 min · technician

PRINT

- Worksheet x24 (2 sheets each, pages 1 and 2 double-sided)
- Fingerprint File x7 (A4 single-sided: 6 groups + 1 display)
- Record cards x24 on A4 white card (2 per sheet, 12 sheets, cut)
- Teacher copy: kit card (A5)
- Extra Fingerprint File x1 to cut for the display glass (optional)

GET OR MAKE (NEW THIS LESSON)

- Soft pencils 2B to 4B x24 (4 per group)
- Clear sticky tape x6 rolls (1 per group)
- A4 white card x12 sheets (record cards)
- Scrap paper x24 sheets (one side used is fine)
- Magnifying glasses x12 (2 per group)
- Trays x6 (1 per group)
- Cloths for wiping tables (number not stated)
- Paper towels or wet wipes (a few per group)
- Visualiser x1 (may already be in the room)
- Class tally table drawn on the board
- Drinking glass with scene print taped on (optional)

Have ready from earlier weeks: Class box, Case files x24 (with suspect cards), Scientist's Logbooks x24, Writing pencils x24, Role cards incl. Language buddy x6 sets, Sentence frame card sets x6, My Science Dictionary x24, Classroom language cards x24

Chinese: draft, needs native-speaker check